

ATROPOS PHASE 20+ ARCHITECTURE + GAP MAP v2

Authority: Phase 20 laws (user-locked) + Source Docs 1–3 + Blueprint Phase 20 + 29 Jul 2026 export + redesigned lakehouse (R2+CAS+structural manifests) + mathematical superiority research

Density: ultra-compressed atoms · fingerprints · predicates · edges · shall · RESEARCH · IMPL (every atom)

Core Engine (0–19) and HOE: separate documents

Non-source superiority primitives: marked [NON-SOURCE]

0. AUTHORITY LEDGER

- Original Source Docs 1–3 = immutable unless explicitly superseded by versioned amendment
- Lakehouse substrate (locked): Cloudflare R2 primary store · CAS key = SHA-256 full digest · structural manifest (header|subheader|prose|code|list|table|warning|example|anti-pattern + byte offsets + parent hash) · local DLOI→CAS→location index · NL prompts auto-retrieve from lakehouse · web/curl only on explicit DAG research step
- Capacity target: 5 GB ≈ 900 M high-signal words · Security = Priority 1 taxonomy
- Existing owners to compose first: AutonomousLearningAdvisor · EvaluationEngine · SelfImprovingCompilationLoop · LocalMemoryStore · EvidenceCollector · AuditorService · VerifiedCompletionGate · SafeJarSwapGate · recovery services · DloiService · OntologicalAddressRouter
- Narrow new atomic files only when no existing owner can hold the responsibility
- Phase 11 = sole mechanical self-build execution engine; Phase 20 only decides
- Three research planes never collapse: (1) lakehouse retrieval on NL prompt (2) application research document → SpecGraph/fallback DAG (3) Phase 20 self-improvement evidence→proposal→amendment

1. PHASE 20 ARCHITECTURE LAWS (20.1–20.20)

LAW	Predicate	GapType	RESEARCH	IMPL
20.1	Original authority immutable unless explicitly superseded	OPEN	RESEARCH: Locate every write path that can touch Source Doc hashes or registry entries; list any silent mutation site.	IMPL: Guard all authority writes behind amendment-only path; original

LAW	Predicate	GapType	RESEARCH	IMPL
				docs read-onl y.
20.2	Runtime observations are evidence candidates, not authority	OPEN	RESEARCH: Trace current observation → action paths; identify any that skip evidence ledger.	IMPL: Force every observat ion through Evidence Record before any further transitio n.
20.3	Evidence may become memory only with provenance + hashes	OPEN	RESEARCH: Inspect LocalMemorySt ore write signatures; confirm provenance fields exist or are missing.	IMPL: Extend memory write to require evidence hash + provena nce; reject bare writes.
20.4	Memory may become proposal only after reproducibility classification	OPEN	RESEARCH: Find any direct memory→actio n or memory→code path that bypasses R(d).	IMPL: Insert Reprodu cibility Gate as mandato ry step before Proposal Generato r.
20.5	No proposal may request source mutation without necessity proof	OPEN	RESEARCH: Define machine-checka ble necessity	IMPL: Reject proposal s

LAW	Predicate	GapType	RESEARCH	IMPL
			schema; list current proposal fields.	missing necessity proof block.
20.6	Every proposal declares baseline, target, guardrails, territory, risk, rollback	OPEN	RESEARCH: Compare existing proposal/advisor structures against the six required fields.	IMPL: Make the six fields mandatory in proposal schema; fail closed.
20.7	Proposing component cannot approve its own proposal	OPEN	RESEARCH: Identify current approval call sites; check for self-approval paths.	IMPL: Enforce proposer-id $\neq$ approver-id in Auditor gate.
20.8	Accepted proposals become versioned authority amendments with independent hashes	OPEN	RESEARCH: Confirm CAS write path for amendments; list missing hash fields.	IMPL: Amendment = new CAS object; never overwrite original authority hash.
20.9	Only accepted amendments may generate self-build atoms/DAG nodes	OPEN	RESEARCH: Trace atom/DAG generation entry points; find any that accept raw proposals.	IMPL: Gate atom compilation on accepted-amendment hash only.

LAW	Predicate	GapType	RESEARCH	IMPL
20.10	Self-build execution uses same bounded engine as external applications	OPEN	RESEARCH: Diff Phase 11 self-build path vs App Factory path; list divergences.	IMPL: Route both through identical Bounded WorkExecutor + territory + verifier.
20.11	Self-build requires stricter attestation, verification, rollback, installed-runtime proof	OPEN	RESEARCH: List extra gates already present in SelfHost* vs ordinary app build.	IMPL: Keep stricter gate set; document as mandatory for self target.
20.12	No self-change may weaken immutable governance or safety invariants	OPEN	RESEARCH: Enumerate invariant list; find any proposal path that can touch them.	IMPL: Meta-level proposals require human-authorized amendment class.
20.13	Every promoted change must prove measurable improvement against predeclared baseline	OPEN	RESEARCH: Locate EvaluationEngine metric hooks; confirm before/after comparison exists.	IMPL: Require I(p) evaluation before promotion; rollback on failure.
20.14	Every promoted change must	OPEN	RESEARCH: Find any	IMPL: Add

LAW	Predicate survive an observation period	GapType	RESEARCH post-promotion cooldown or observation timer.	IMPL mandatory observation-period state before next change in same subsystem.
20.15	Failed improvement claims produce evidence and learning, not silent retries	OPEN	RESEARCH: Trace failure handling in current learning/advisor paths.	IMPL: Failed I(p) writes evidence + learning ledger entry; no silent retry.
20.16	Repeated failed proposals are quarantined	OPEN	RESEARCH: Check for existing quarantine or retry-limit structures.	IMPL: After N failures, mark proposal quarantined; require new evidence to reopen.
20.17	Self-improvement loop is rate-limited, budgeted, and depth-bounded	OPEN	RESEARCH: Inventory current rate/budget controls on autonomous loops.	IMPL: Enforce max depth, max proposals/period, max files/LOC,

LAW	Predicate	GapType	RESEARCH	IMPL
				token/compute budget.
20.18	Restart continuity restores evidence, proposal, authority, territory, and next action	OPEN	RESEARCH: Map CrashRecoveryService / StateSnapshot fields against required restore set.	IMPL: Extend restore to include proposal + amendment + territory + next action.
20.19	Completion claims cite authority, amendment, implementation, evidence, and runtime hashes	OPEN	RESEARCH: Inspect current completion claim schema; list missing hash citations.	IMPL: Completion claim must embed all five hash classes or is rejected.
20.20	System must explain why/where/how it changed and why the result is better	OPEN	RESEARCH: Locate any existing explanation/evidence browser for self-changes.	IMPL: Proof bundle must be queryable and human-readable post-promotion.

2. DETERMINISTIC SELF-IMPROVEMENT LOOP

ATOM-ID	Transition	AcceptancePredicate	GapType	RESEARCH	IMP
P20-L01	Runtime Observation $n \rightarrow$	observation carries ID, timestamp, runtime/project/goal	OPEN	RESEARCH: List all current	IMP: L: Nor

ATOM-ID	Transition Evidence Record	AcceptancePredicate /node IDs, authority+env fingerprints, exit code, bounded output, artifact hashes, frequency, severity	GapType	RESEARCH: observation emission sites; check field completeness.	IMPL: malize observation scheme; reject incomplete observations.
P20-L02	Evidence → Learning Candidate	evidence survives provenance + hash checks	OPEN	RESEARCH: Trace Evidence Collector → memory /advisor handoff.	IMPL: Only hashed, provenance-bearing evidence becomes Learning Candidate.
P20-L03	Learning Candidate → Reproducibility Gate	R(d) holds	OPEN	RESEARCH: Formalize R(d) against existing failure signatures.	IMPL: Implement R(d) as pure func

ATOM-ID	Transition	AcceptancePredicate	GapType	RESEARCH	IMPLEMENTATION over evidence records.
P20-L04	Reproducibility → Deficiency Classification	advances only on deterministic reproduce, frequency threshold, single invariant break, safety-critical, or blocked requirement	OPEN	RESEARCH: Define frequency thresholds and safety-critical classes.	IMPLEMENT: Classifier emits typed deficiency or drops candidate.
P20-L05	Deficiency → Improvement Proposal	proposal contains necessity proof + six mandatory fields	OPEN	RESEARCH: Compare Autonomous Learning Advisor output to required proposal schema.	IMPLEMENT: Propose Generator or emits only complete proposals.

ATOM-ID	Transition	AcceptancePredicate	GapType	RESEARCH	IMP
P20-L06	Proposal → Expected-Value Proof	metrics declared before mutation; I(p) defined	OPEN	RESEARCH: Locate metric declaration sites in EvaluationEngine.	IMP L: Proposal without predicted declared metric space is rejected.
P20-L07	Proposal → Risk/Territory Proof	impact closure computed; territory bounded	OPEN	RESEARCH: Reuse Territory Service / impact analysis if present.	IMP L: Attach computed territory + impact closure to proposal.
P20-L08	Proposal → Auditor Acceptance	proposer ≠ approver	OPEN	RESEARCH: Map AuditorService decision points.	IMP L: Auditor gate enforces identity separation

ATOM-ID	Transition	AcceptancePredicate	GapType	RESEARCH	IMPLEMENT
					n + anti-gaming checks.
P20-L09	Acceptance → Versioned Authority Amendment	amendment is CAS object with own hash + evidence + auditor hashes	OPEN	RESEARCH: Confirm CAS write path and structural manifest support.	IMPLEMENT: Write amendment as new CAS object; never mutate original.
P20-L10	Amendment → Atom/DAG Compilation	only affected atoms + dependents	OPEN	RESEARCH: Find current atom/DAG compilation entry; measure blast radius.	IMPLEMENT: Impact-closure compiler emits bounded DAG only.
P20-L11	DAG → Phase 11	Phase 11 runs under strict territory	OPEN	RESEARCH: Confirm SelfHost	IMPLEMENT: Phase 20

ATOM-ID	Transition Self-Build Execution	AcceptancePredicate	GapType	RESEARCH H / Bounded WorkExecutor entry from external contract.	IMP L emit s cont ract; Phase 11 exec utes ; no dire ct sour ce edit fro m Phase 20.
P20-L1 2	Execution → Independent Verification	VerifiedCompletionGate + no self-verification	OPEN	RESEARCH: Verify every self-build path already hits VerifiedCompletionGate.	IMP L: Non zero exit or miss ing evid ence bloc ks VER IFIED.
P20-L1 3	Verification → Before/After Evaluation	I(p) evaluated	OPEN	RESEARCH: Hook EvaluationEngine before/after comparis on.	IMP L: Com pute I(p); emit pass /fail with met

ATOM-ID	Transition	AcceptancePredicate	GapType	RESEARCH	IMPLEMENT
P20-L14	Evaluation → Promotion or Rollback	improve $\wedge$ guardrails → promote; else rollback	OPEN	RESEARCH: Locate SafeJarSwapGate / rollback paths.	IMPLEMENT: Promotion only on I(p) success; else automatic rollback.
P20-L15	Outcome → Learning Ledger Update	success or failure both durable	OPEN	RESEARCH: Confirm learning ledger write on both paths.	IMPLEMENT: Both promotion and rollback write CAS evidence + ledger entry.

3. LAKEHOUSE-BACKED LEDGERS

ATOM-ID	Ledger	Predicate	GapType	RESEARCH	IMPL
P20-LH01	Evidence Ledger	every run produces CAS-hashed evidence with structural manifest; DLOI-addressable	OPEN	RESEARCH: Map EvidenceCollector output to CAS + manifest schema.	IMPL: All evidence written as CAS objects with manifests.
P20-LH02	Memory / Learning Ledger	evidence → memory only with provenance + hashes	OPEN	RESEARCH: Inspect LocalMemoryStore for provenance fields.	IMPL: Memory write requires evidence hash; bare writes rejected.
P20-LH03	Proposal Store	proposals are CAS objects with full necessity + metric + territory package	OPEN	RESEARCH: Define proposal CAS schema; check existing advisor output.	IMPL: Store proposals as CAS; index by DLOI.
P20-LH04	Accepted Amendment Registry	independent hash; original Source Docs untouched	OPEN	RESEARCH: Design registry as CAS + DLOI; forbid overwrite of original hashes.	IMPL: Registry append-only for amendments.

ATOM-ID	Ledger	Predicate	GapType	RESEARCH	IMPL
P20-LH05	Structural Manifest Discipline	every object carries region-type + byte-offset + parent-hash manifest	OPEN	RESEARCH: Specify manifest format matching lakehouse taxonomy.	IMPL: Manifest written atomically with every CAS put.
P20-LH06	NL Auto-Retrieve Rule	ordinary NL → DLOI lookup only; web/curl only on explicit DAG research step	OPEN	RESEARCH: Locate prompt → context assembly path; insert lakehouse retrieve.	IMPL: Default retrieve from lakehouse; network gated by DAG flag.
P20-LH07	Retrieve →Generate→Verify Sandwich	same sandwich for user apps and self-improvement (stricter on self)	OPEN	RESEARCH: Confirm existing retrieve/generate/verify stages.	IMPL: Self-improvement path reuses sandwich with extra gates.

4. GOVERNANCE LEARNING CANDIDATE SET

ATOM-ID	Candidate	Evidence Example	GapType	RESEARCH	IMPL
P20-G01	False VERIFIED	Antigravity OOM → claimed Verified	OPEN	RESEARCH: H:	IMPL:

ATOM-ID	Candidate after nonzero compile/test exit	Evidence Example	GapType	RESEARCH: Instrument completion reporters for exit-code vs claimed state.	IMPLEMENTATION: First amendment: nonzero exit forbids VERIFIED.
P20-G02	Self-verification / self-approval	component approves its own proposal	OPEN	RESEARCH: Audit all approval call sites for identity equality.	IMPLEMENTATION: Hard reject when proposer-id == approver-id.
P20-G03	Out-of-territory mutation attempt	write outside assigned territory	OPEN	RESEARCH: Confirm Territory Enforcer coverage on all mutation paths.	IMPLEMENTATION: Any out-of-territory attempt becomes governance evidence.

ATOM-ID	Candidate	Evidence Example	GapType	RESEARCH	IMPLEMENTATION
P20-G04	Secret leakage or redaction failure	raw secret in log/transcript/evidence	OPEN	RESEARCH: Scan redaction filter coverage on evidence/ledger writes.	IMPLEMENTATION: Leak = safety hard-fail; blocks promotion.
P20-G05	Context / attestation drift	provider returns wrong identity/task/hash	OPEN	RESEARCH: Map ContextAttestationService failure modes.	IMPLEMENTATION: Drift events feed governance ledger.
P20-G06	Evidence omission on completion claim	VERIFIED without required evidence hashes	OPEN	RESEARCH: List required evidence set for VERIFIED.	IMPLEMENTATION: Missing evidence → claim rejected.
P20-G07	Authority-hash mismatch / silent Source Document mutation	original hash changed without amendment	OPEN	RESEARCH: Add continuous hash monitor on Source	IMPLEMENTATION: Mismatch = immediate

ATOM-ID	Candidate	Evidence Example	GapType	RESEARCH Doc registry.	IMPLEMENTATION candidate.
P20-G08	Oscillating self-changes without observation period	repeated flip-flop on same subsystem	OPEN	RESEARCH: Detect oscillation patterns in promotion history.	IMPLEMENTATION: Oscillation triggers quarantine + cooldown.
P20-G09	Completion-state vocabulary collapse	IMPLEMENTED/COMPILED/TESTED/VERIFIED/BLOCKED not distinguished	OPEN	RESEARCH: Enumerate current status strings.	IMPLEMENTATION: Enforce five-state vocabulary in all reporters.

5. HARD BOUNDARIES + ANTI-RECURSION

ATOM-ID	Control	Predicate	GapType	RESEARCH	IMPLEMENTATION
P20-H01	Immutable invariants	human authority · source immutability · territory ·	OPEN	RESEARCH: Produce canonical invariant list from	IMPLEMENTATION: Any proposal touches

ATOM-ID	Control	Predicate secrets · paid lock · independent verification · evidence · rollback · acceptance · anti-gaming · attestation · safety hard-fails	GapType	RESEARCH Source Docs + Blueprint.	IMPL ng these requir es huma n-aut horize d class.
P20-H02	Meta-level separation	proposal that rewrites verification/t erritory/imp rovement predicates needs human gate	OPEN	RESEARCH: Classify which proposals are meta-level.	IMPL: Meta-l evel flag forces extern al autho rization.
P20-H03	Rate / depth / budget limits	max depth, max proposals/pe riod, max files/LOC, max retries, token/compu te budget	OPEN	RESEARCH: Inventory existing autonomou s rate limits.	IMPL: Enforc e nume ric bound s; fail closed when excee ded.
P20-H04	Cooldown + observatio n period	after promotion, observe before next change in same subsystem	OPEN	RESEARCH: Find any existing cooldown timers.	IMPL: Mand atory obser vation -perio d state machi ne.

ATOM-ID	Control	Predicate	GapType	RESEARCH	IMPL
P20-H05	Quarantine	repeated failed proposals quarantined	OPEN	RESEARCH: Design quarantine record schema.	IMPL: N failures $\rightarrow$ quarantined; new evidence required to reopen.
P20-H06	Termination ranking	lexicographic potential $\rightarrow$ 0 stops loop	OPEN	RESEARCH: Define potential components (deficiencies, age, cooldown, budget).	IMPL: Potential evaluated each cycle; zero terminates.

6. [NON-SOURCE] MATHEMATICAL SUPERIORITY PRIMITIVES

ATOM-ID	Primitive	Predicate	GapType	RESEARCH	IMPL
P20-NS01	Proof-carrying amendments	amendment CAS object contains deficiency hashes + necessity + metric space + auditor hash + impact closure	OPEN	RESEARCH: Design proof-bundle schema as CAS object.	IMPL: Acceptance requires complete proof bundle.
P20-NS02	Formal reproducibility predicate $R(d)$	machine-checkable; noise cannot generate proposals	OPEN	RESEARCH: Encode $R(d)$ against	IMPL: Pure function;

ATOM-ID	Primitive	Predicate	GapType	RESEARCH	IMPL
P20-NS03	Metric-space improvement $I(p)$	distance to target decreases $\wedge$ guardrails preserved	OPEN	RESEARCH: Define metric space interface in Evaluation Engine.	unit-tested. IMPL: $I(p)$ computed before promotion; failure $\rightarrow$ rollback.
P20-NS04	Termination ranking function	potential guarantees no infinite cycle	OPEN	RESEARCH: Choose lexicographic components.	IMPL: Potential evaluated every loop iteration.
P20-NS05	Object/meta-level separation	system cannot rewrite its own success predicates without external gate	OPEN	RESEARCH: Tag all predicates as object-level or meta-level.	IMPL: Meta-level changes require human authorization class.
P20-NS06	Lattice of proposals	contradictory proposals incomparable; only maximal consistent	OPEN	RESEARCH: Define conflict relation on proposals.	IMPL: Audit or promotes only consis

ATOM-ID	Primitive	Predicate sets accepted	GapType	RESEARCH	IMPL tent maxi mal sets.
P20-NS07	Unified CAS substrate	evidence, memory, proposals, amendments, proof bundles all CAS + manifests	OPEN	RESEARCH: Confirm single CAS root can hold all five kinds.	IMPL: One CAS + DLOI index; no parallel stores.

7. CONTINUOUS POST-100% SUPERIORITY

ATOM-ID	Predicate	GapType	RESEARCH	IMPL
P20-S01	Competitive observation ledger	OPEN	RESEARCH: Define schema for external agent/CLI/swarm behavior observations.	IMPL: Observations enter evidence ledger under competitive tag.
P20-S02	Superiority proposals from competitive observations	OPEN	RESEARCH: Reuse same proposal pipeline with competitive source tag.	IMPL: Same gates; no special bypass.
P20-S03	Accepted superiority amendments follow identical hash + Phase 11 + before/after path	OPEN	RESEARCH: Confirm no parallel promotion path exists.	IMPL: Single promotion path for internal and competitive

ATOM-ID	Predicate	GapType	RESEARCH	IMPL
P20-S04	Ongoing metric dashboard	OPEN	RESEARCH: List required metrics (false-VERIFIED rate, territory violations, recovery completeness, tokens-per-verified-change, observation success).	amendments. IMPL: Dashboard reads from evaluation + learning ledgers.

8. ORDERED GAP DAG

P20-LH01...LH07 → P20-L01...L15 → P20-G01...G09 → P20-H01...H06 → Laws 20.1–20.20 green → P20-NS01...NS07 → P20-S01...S04 → 100% PHASE 20 GATE GREEN

Phase 11 remains the only component allowed to mutate ATROPOS source. Phase 20 only emits accepted, hash-pinned contracts.

9. RAW CLOSURE METRICS

- Laws 20.1–20.20: all OPEN (architecture locked, implementation pending)
- Loop transitions: all OPEN
- Lakehouse ledgers: OPEN (substrate design locked)
- Governance candidates: OPEN (Antigravity OOM = first canonical)
- Hard boundaries + anti-recursion: OPEN
- [NON-SOURCE] superiority: all OPEN
- Continuous post-100%: OPEN

Every atom carries RESEARCH (ingesting LLM prompt) and IMPL (necessary implementation note).

10. HANDOFF BLOCK

AUTHORITY: Phase 20 user-locked architecture 2026-07-29 + redesigned lakehouse (R2+CAS+structural manifests) + three research-plane separation.
NON-GOALS: Do not silently edit Source Docs 1–3; do not let observation

become law; do not allow proposer to approve itself; do not collapse user-app research into self-improvement law; do not create a second storage root.

RESEARCH PLANES (never collapse):

1. NL prompt → automatic lakehouse DLOI retrieval
2. Insufficient knowledge → research document → SpecGraph / fallback DAG
3. Runtime deficiency → Phase 20 evidence → proposal → amendment

IMPLEMENTATION ORDER: lakehouse ledgers → loop transitions → governance candidates → hard boundaries → laws green → superiority primitives → continuous superiority.

DELIVERABLE LANGUAGE: shall/must/evidence/acceptance; path fingerprints; update only changed delta-register rows.

FIRST CANONICAL AMENDMENT TARGET: nonzero compile/test exit forbids VERIFIED (from Antigravity OOM evidence).

Every atom above includes an ingesting RESEARCH prompt for the next LLM and a concrete IMPL note.

End of Phase 20+ Architecture + Gap Map v2.

Core Engine (0–19) and HOE remain separate documents.